Per-transfer sunset hardening: the endpoint joint is no longer thin (S = x1.129 -> x2.994 realized)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

The B5 T6-conditional hypothesis H-ENDPOINT-THINNESS-ACCEPTED exists because the certified endpoint joint saturates at the sunset cap $S = \times 1.13$. This note attacks that cap with the methodology that certified the quartic channel: replace a sup-bound kernel by its realized per-transfer evaluation. **Scope:** the sunset kernel factor of the endpoint bookkeeping, at the endpoint dressing, on the admissible chord set. **Not claimed:** any gate or tier change; any new analytic theorem (this is an EXECUTED evaluation of the existing bound's kernel, certificate-candidate grade).

2. Pre-registered gates (set before computation)

G1: $\text{shape}_{\max} = \max_{t \in \text{chords}} J_{\text{eff}}(t)/J_0 \geq 1 \Rightarrow$ hardening FAILS (honest negative). G2: $S_{\text{realized}} \leq 1.2 \Rightarrow$ immaterial. G3: success $\Rightarrow$ report S_{realized} and the recomputed joints at $\rho_{\text{lat}} = 6.55$ and 12.6. **Outcome: G3.**

3. The replacement and why it is exact

The anchor bound (scscope-mendpoint-evaluation v1.1) is $b = \frac{1}{2} u_{\text{eff}}^2 (2I)\, 6\, J_0 M$ with $J_0 = \sup_t J(t)$ at the 4×10^{-4} dressing: the 3-propagator sunset kernel $(J * G)(t)$ (bubble × propagator) bounded by $\sup J \cdot \int G$. The realized replacement evaluates the convolution instead of bounding it. **Derivation compatibility (explicit):** the existing bookkeeping multiplies the TOTAL transfer weight by ONE kernel constant; ANY constant that upper-bounds the kernel on every realized transfer is admissible there, and $\max_{t \in \text{chords}} J_{\text{eff}}(t)$ is such a constant (machine-asserted $J_{\text{eff}}(t) \leq J_0$ pointwise and every admissible transfer lies in the chord set) – so the replacement is a strictly valid sharpening of the same inequality, not a new bookkeeping. The kernel is:

$$J_{\text{eff}}(t) = \frac{N(t)}{N_0}, \qquad N(t) = \iint \frac{k^2}{D(k)}\, J\left(\sqrt{k^2 + t^2 + 2kt\mu}\right) d\mu\, dk,$$

$$N_0 = \iint \frac{k^2}{D(k)}\, d\mu\, dk, \qquad D(k) = \hat{r}_{\text{end}} + C(k^2 - q_0^2)^2,$$

with ALL propagators at the endpoint dressing $\hat{r}_{\text{end}} = 0.33675$. The absolute normalisation cancels in the ratio: no $(2\pi)^3$ convention enters (the factor-2 error class is structurally excluded), and $J_{\text{eff}}(t) \leq J_0$ pointwise (machine assert). Admissible transfers obey $|t| \geq t_{\min} = 2q_0 \sin(\theta_{\min}/2) = 0.420$, so the chord set $[t_{\min}, 2q_0]$ covers every realized transfer.

4. Results (scscope_sunset_pertransfer.py, 8/8 asserts PASS)

Calibration: the anchor is REPRODUCED, $S_{\mathrm{anchor}} = \times 1.129$ ($J_0 = 0.2896$, $M_{\mathrm{END}} = 0.104953$). Two-way pin: $J_{\mathrm{eff}}(t \to 0)$ by the convolution and by the degenerate 1D integral agree to 10^{-5}. Grid-doubling drift on the worst chord: 0.00%.

$$\mathrm{shape}_{\max} = \frac{\max_t J_{\mathrm{eff}}(t)}{J_0} = \frac{0.1093}{0.2896} = 0.377 \quad (\text{at } t = t_{\min} = 0.420),$$

$$S_{\mathrm{realized}} = \frac{1.129}{0.377} = \times 2.994.$$

Recomputed joints (certified $R_{\max} = 0.385$ from the quartic certificate run artefact):

$$\mathrm{joint}(\rho{=}6.55):\ \times 1.040 \to \times 2.023, \qquad \mathrm{joint}(\rho{=}12.6):\ \times 1.082 \to \times 2.396,$$

$$\mathrm{cap}(\rho \to \infty):\ \times 1.13 \to \times 2.994.$$

The doubly-conservative structure of the anchor (sup over transfer; loosest dressing) accounts for the factor: the loop average of J at $|t| \geq 0.420$ sits far below the $t \to 0$ sup, exactly as the quartic's Young-vs-realized gap ($1.019 \to 0.385$) did.

5. Consequence for the B5 T7 path (no flip enacted)

If the operator accepts this evaluation as the sunset accounting, the endpoint closure is COMFORTABLE (×2.0–×2.4), and H-ENDPOINT-THINNESS-ACCEPTED – one of the two $\mathcal{H}_{\mathrm{B5}}^{\mathrm{T6}}$ hypotheses – loses its load: its removal would leave H-NONLATTICE-REMAINDER-EXCLUDED (T-030) as the single remaining B5 T7 blocker. This note enacts NOTHING: the acceptance, the GATES.md update, the W_SC re-certification with the hardened S, and any SC-SCOPE package re-issue are operator decisions upon review.

5b. Mixed-dressing justification (operator-required for the final stamp)

Monotonicity argument. Both kernel factors move the SAME way with the dressing: $J(t;r)$ decreases in r (heavier mass, smaller bubble – the direction already certified by the mendpoint table $J(\hat{r}(I)) \leq J_0$), and the loop weight $1/D(k;r)$ decreases in r. Hence over ALL assignments of dressings to the sunset's propagators, the LIGHTEST (anchor, 4×10^{-4}) assignment maximises J_{eff}: it is the adversarial corner, and matched endpoint dressing is the physically correct (per-pattern self-consistent) choice, used non-load-bearingly.

Machine worst-case (script v1.1.0, asserts 9–10). All four (J-grid, D) dressing assignments evaluated on the chords: worst J_{eff} per variant = 0.1093/0.1211/0.1110/0.1230 (end-end / anchor-end / end-anchor / anchor-anchor), confirming the monotone ordering. At the adversarial corner: $\mathrm{shape}_{\mathrm{worst}} = 0.4247$, $S_{\mathrm{worst}} = \times 2.659 > 2$, $\mathrm{joint}(\rho{=}6.55) = \times 1.886$ — the hardening and the gate-removal claim SURVIVE the worst dressing assignment.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *"The weighted-average replacement may be doing bookkeeping work the bound's derivation does not permit (the sum over transfers might need the kernel at EACH transfer, not a sup)."* ADDRESSED: the original bound already uses ONE kernel value (the sup) times the total weight; replacing the global sup by the sup over the ADMISSIBLE chord set is a strictly valid intermediate ($J_{\mathrm{eff}}(t) \leq \max_{\mathrm{chords}} J_{\mathrm{eff}} \leq J_0$ pointwise on every realized transfer); no per-transfer redistribution is claimed.

(β) *"The dressing choice (all propagators at $\hat{r}_{\text{end}}$) could be optimistic."* RESOLVED in v1.2 (operator-required): Sec. 5b gives the monotonicity argument AND the machine worst-case – even the adversarial (anchor, anchor) assignment leaves $S_{\text{worst}} = \times 2.659 > 2$ and joint $= \times 1.886$; the matched endpoint choice is not load-bearing.

(γ) *"J_{eff} peaks at $t_{\min}$ – a competitor just above the admissibility floor gets the weakest hardening; is the floor itself certified?"* DISMISSED: $t_{\min}$ comes from $\theta_{\min}(\hat{r})$, the SAME admissibility input the whole B5 chain uses (machine-asserted here); at $t_{\min}$ the hardening is already $\times 2.65$ relative to J_0, and the gate G3 threshold was pre-registered well below the observed value.

Quantitative sanity check (mandatory). Direction: a smaller kernel can only RAISE the sunset margin – selection-favorable by construction, and the mechanism (loop average < sup) is fixed by $J(t)$ monotonicity on the chord range, not tuning. Limit case: at $t \to 0$ the two-way pin agrees to 10^{-5}; at the anchor inputs the script REPRODUCES $\times 1.129$ before any replacement. Magnitude: shape$_{\max} = 0.377$ is comparable to the quartic's realized/Young ratio ($0.385/1.019 = 0.378$) – the same kernel geometry drives both, an independent cross-consistency.

7. Result footer

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Result ID:               B5-BEYOND-LAYER-BOUND / SC-SCOPE sunset
                         per-transfer hardening (B5 T7 path)
Precise statement:       EXECUTED (8/8): replacing the single-J0 sup
                         anchor by the realized D-weighted loop
                         average J_eff(t) on the admissible chords
                         (all propagators at the endpoint dressing)
                         gives shape_max = 0.377, S = x1.129 ->
                         S_realized = x2.994; endpoint joint x1.040
                         -> x2.023 (rho=6.55), x1.082 -> x2.396
                         (rho=12.6); saturation cap x1.13 -> x2.994.
                         Gate G3-SUCCESS (pre-registered).
Scope:                   the sunset kernel factor at the endpoint, on
                         the admissible chord set. Certificate-
                         candidate (EXECUTED evaluation); no new
                         analytic theorem; no flip.
Evidence grade:          EXECUTED certificate-candidate (T4-grade
                         evidence for the operator decision).
Reproduction command:    python
                         codes/vacuum/scscope_sunset_pertransfer.py
Falsifier:               a realized admissible transfer with
                         J_eff(t) > max_chords J_eff (chord-set
                         escape); anchor non-reproduction; grid
                         drift > 1%.
Tier before / after:     CONFIRMED + ENACTED (operator 2026-06-12):
                         H-ENDPOINT-THINNESS-ACCEPTED REMOVED; B5
                         stays T6-conditional with the single
                         hypothesis H-NONLATTICE-REMAINDER-EXCLUDED;
                         registered SC-SCOPE-SunsetHardened-T6-260612.
No-overclaim statement:  hardening of the KERNEL FACTOR only, within
                         the existing bookkeeping (derivation-
                         compatibility stated explicitly, Sec. 3);
                         the mixed-dressing audit is DELIVERED
                         (Sec. 5b, worst case S = x2.659); W_SC
                         re-certification with the hardened S is
                         not done here.
Next required action:    operator ratification of the v1.2 mixed-
                         dressing addendum (final gate stamp) ->
                         THEN package SC-SCOPE-SunsetHardened-T6-
                         260612 (sec.14). Mainline: T-030 = the
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single remaining B5 T7 blocker.